



telecommunication

Project



Summary

SyriaTel, a leading telecommunications provider, is seeking a predictive solution to help reduce customer churn. The company has tasked me with developing a model that can accurately forecast whether a customer is likely to stay with the service or leave. By identifying patterns in customer behavior, this model will enable SyriaTel to take proactive measures to retain valuable customers and minimize revenue loss due to churn

Objectives

- **Identify the most effective model for accurately predicting customer churn.**
- **Determine the key features that significantly influence a customer's decision to remain with or leave SyriaTel.**

Best Model

➤ Model: Decision Tree

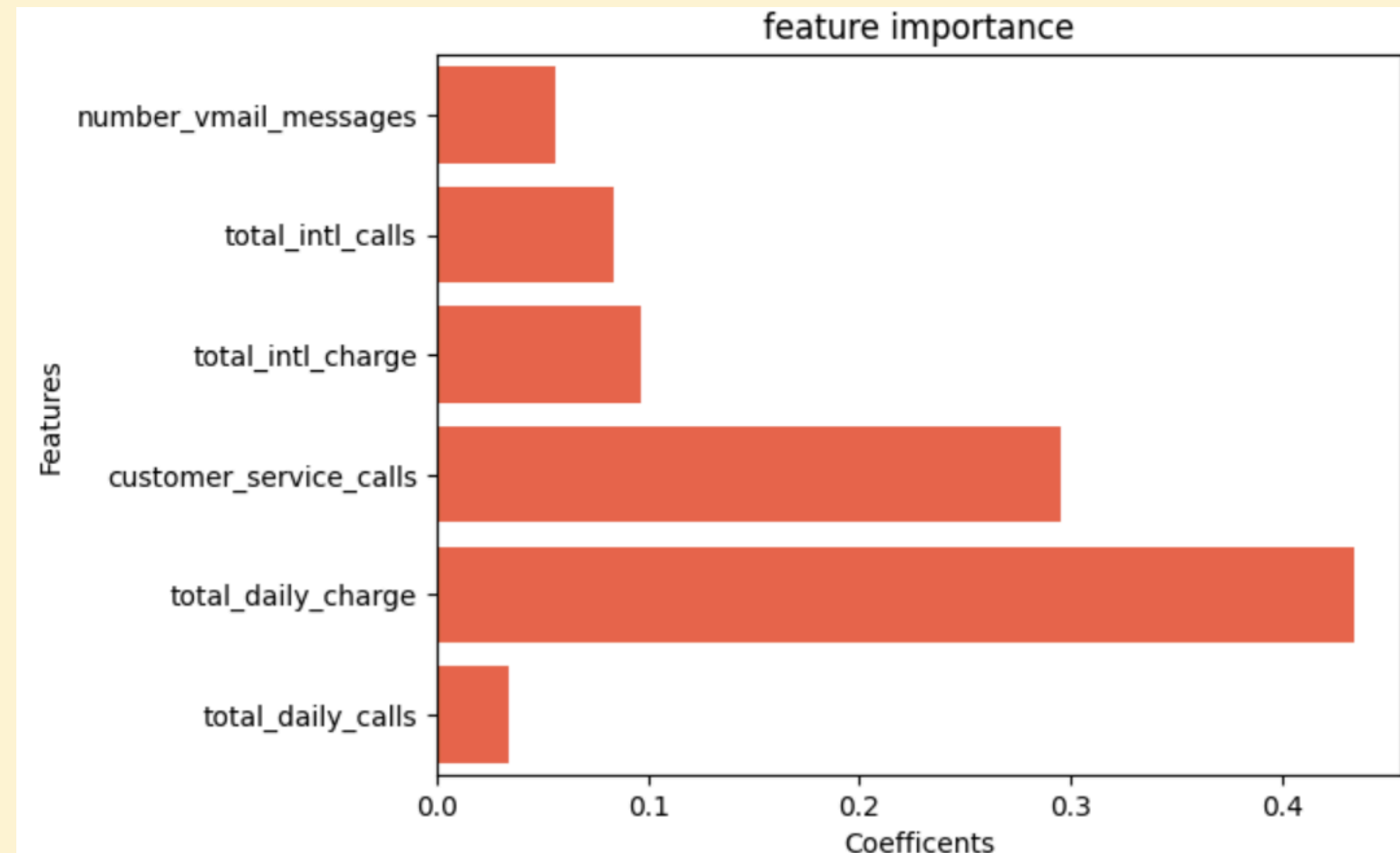
The decision tree model performs well overall, correctly predicting outcomes about 87% of the time. It's particularly good at identifying most of the true positive cases (recall: 78%), though some false positives still occur (precision: 54%). The AUC score of 83% indicates strong overall ability to distinguish between classes, showing that the model has reliable predictive power

Feature Importance

Sure! Here's a simplified, bullet-point version:

- **Main drivers of churn:**
 - Higher daily charges
 - More customer service calls
- **Moderate factors:**
 - International charges and international calls
 - Number of voicemail messages
- **Low impact:**
 - Total daily calls
- **Overall insight:**
 - Customers who spend more or contact support often are more likely to leave.
 - Focusing on pricing fairness and better customer service can help reduce churn.

Feature Importance Visualization



Q&A SESSION



THANK YOU